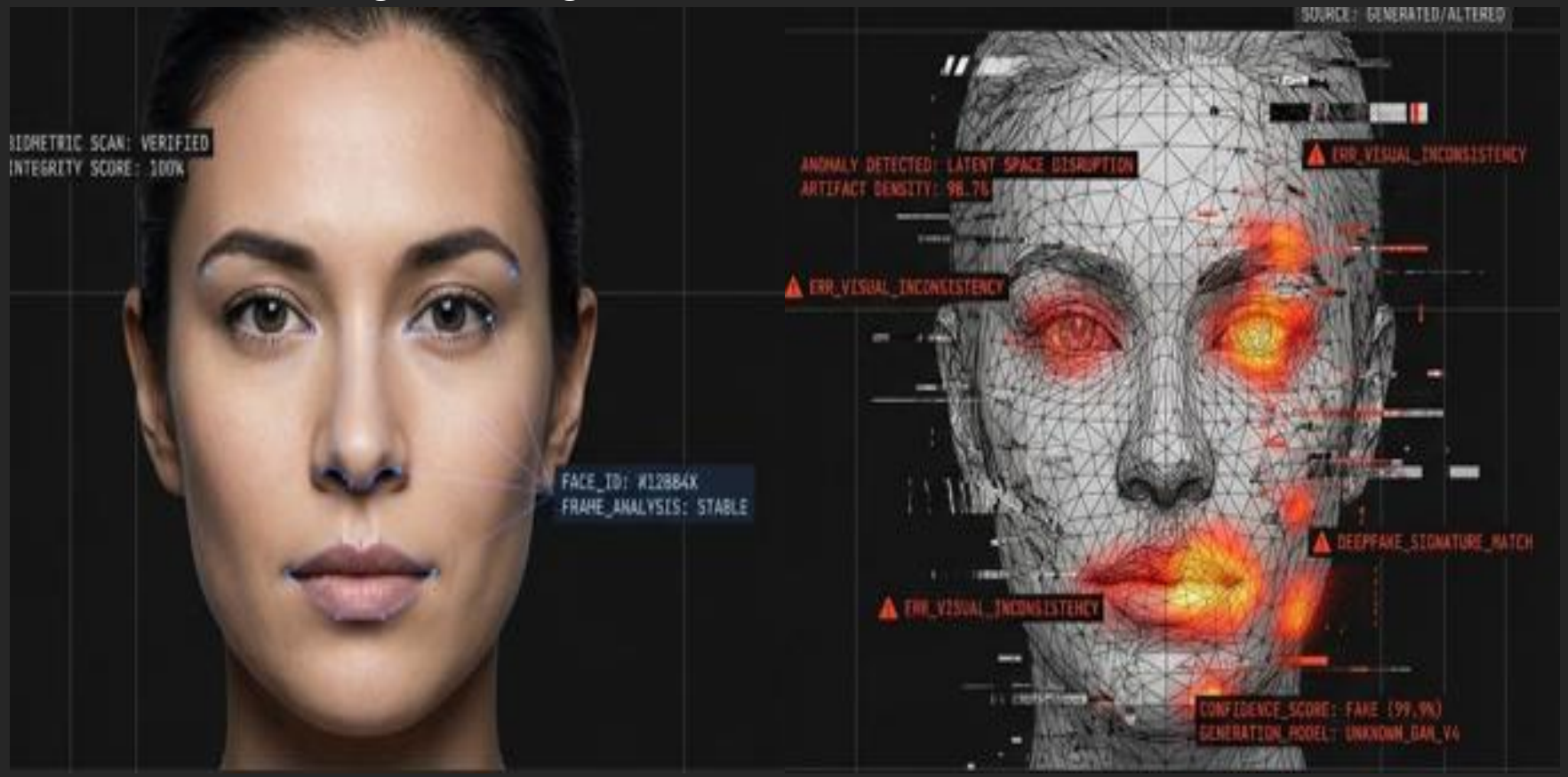
RealVision

Deepfake Detection and Explainability

A Multimodal Investigation using FaceForensics++



Presented by: Menalu Chekol, Odelya Datski and Zohar Shalom

Course: Deep Learning

Lecturer : Idan Tobis

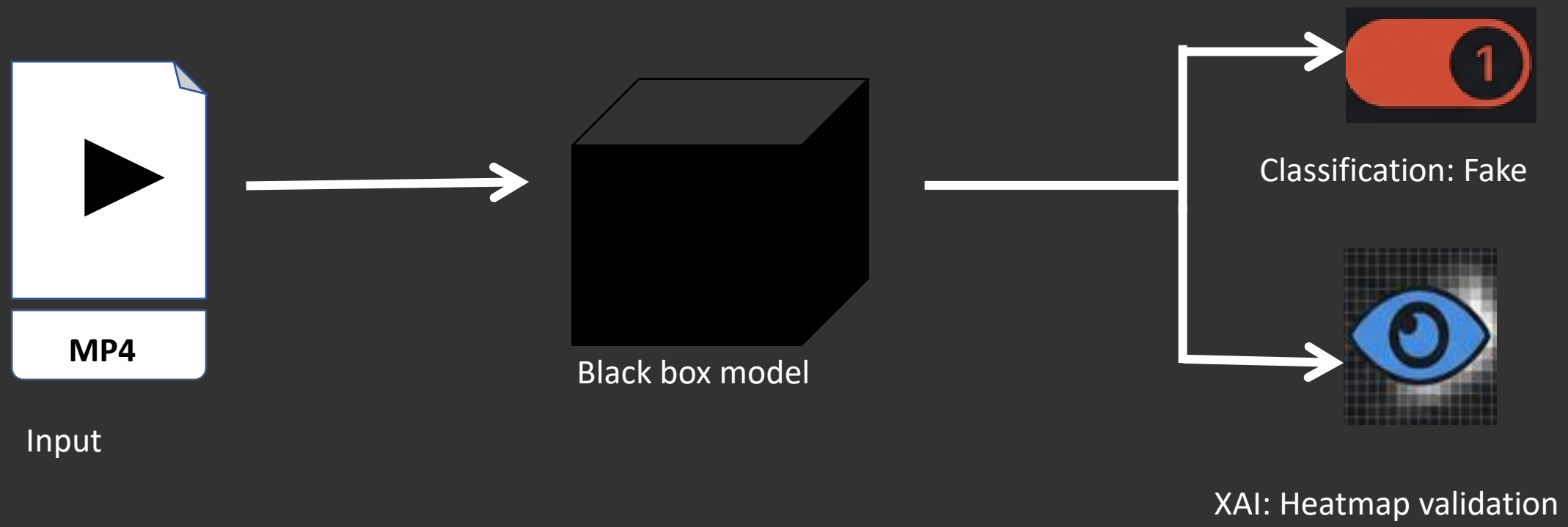
Motivation: Why Deepfake Detection Matters

- **Misinformation**
  Deepfakes spread false content

- **Identity Impersonation**
  Fraud and reputational damage

- **Trust and Security**
  Need reliable, explainable detection

The threat is invisible.
The defense must be explainable



Task: Binary Classification (0: Real / 1: Fake)
Challenge: Manipulated media is indistinguishable from reality.
Goal: Pierce the "Black Box" to understand the decision.

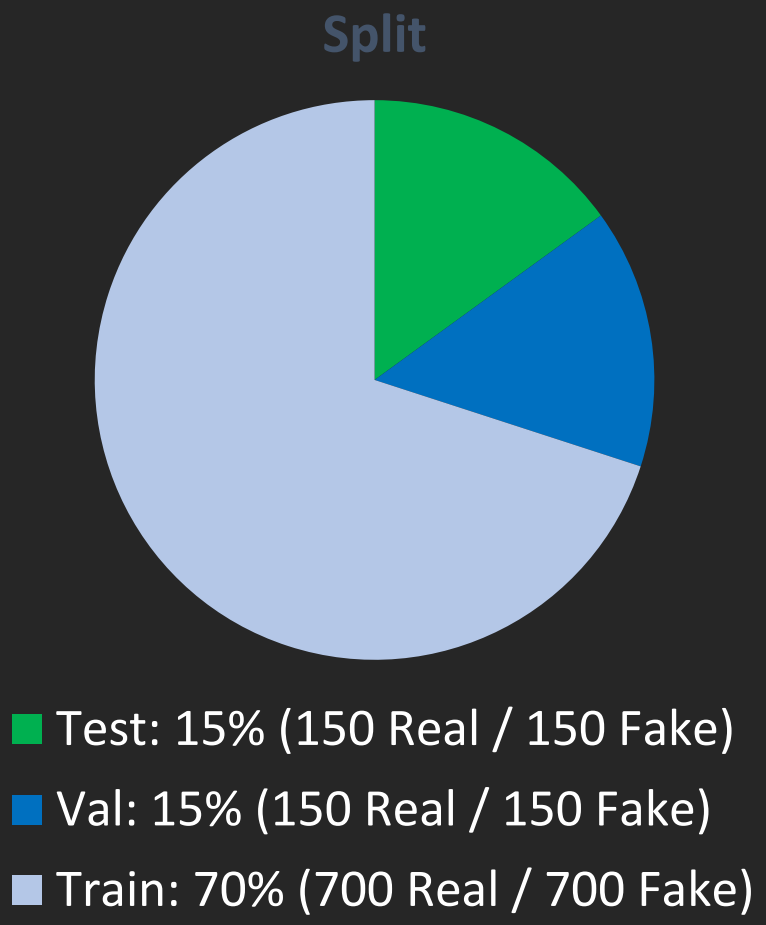
The Evidence: FaceForensics++

Source:
2,000 Balanced Videos

Classes:
1,000 Original / 1,000 Manipulated

Manipulation Types:
Deepfakes, Face2Face, FaceSwap

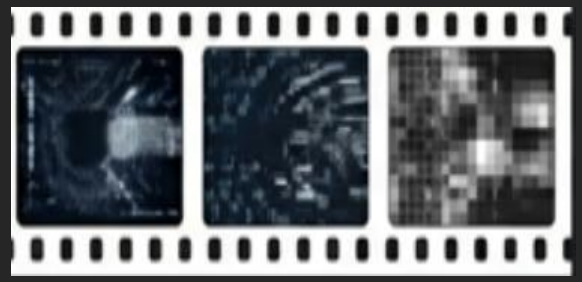
Methodology:
Stratified Split (Preserving Class Balance)



Processing the Evidence

The Multimodal Pipeline

Stream A: Video Forensics



16 Frames (Uniform Sampling) → Resize (224×224) → Tensor (T, 3, H, W) → Augmentation: Horizontal Flip (p=0.5)

Stream B: Audio Forensics

4-Second Clip (ffmpeg) → 16kHz Mono → MFCC Extraction (40 coeffs) → Safety: Zero-Padding

The Baseline

Video-Only Architecture

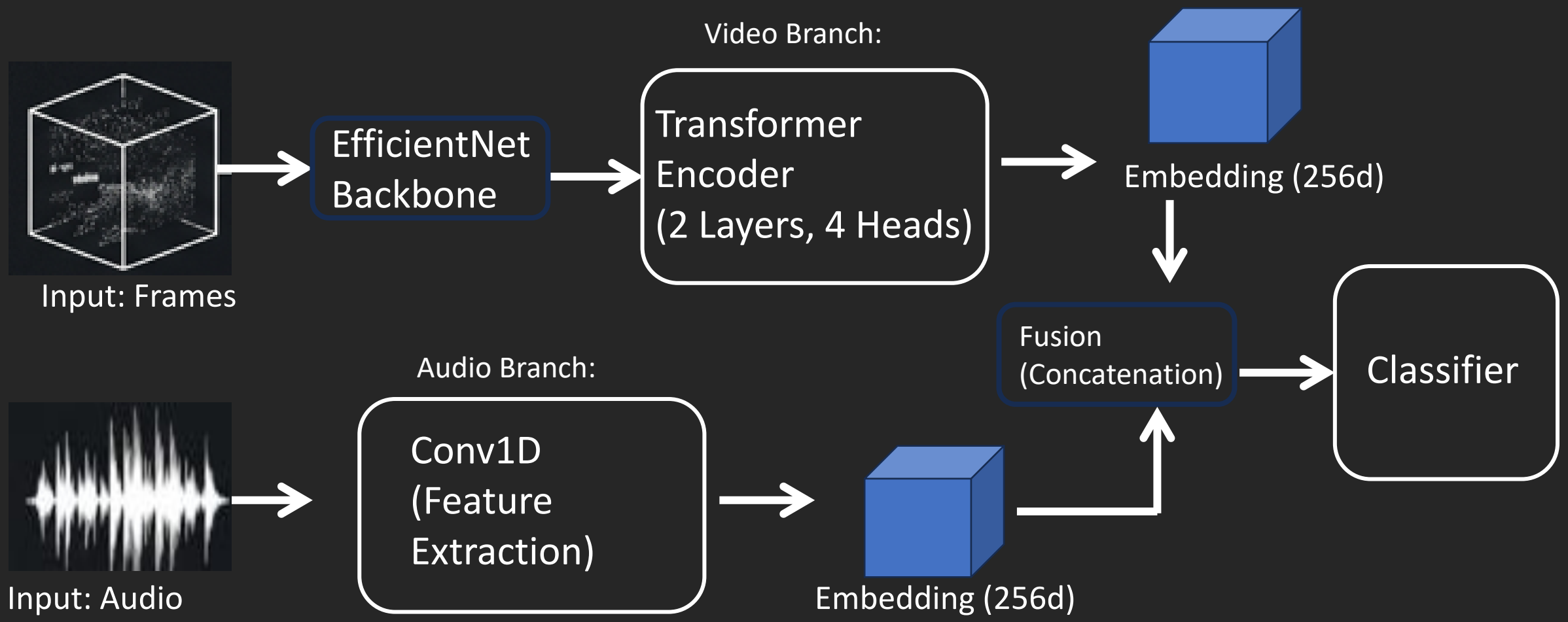
→ **Backbone: EfficientNet-B0 (Pretrained)** → **Pooling (Temporal Average)** → **Linear Classifier**

Control Group:
Visual-only features.

Process:
Frame-wise feature extraction.

The Challenger: RealVision AV

Multimodal Architecture



Training Protocols

Optimizer
AdamW
Adaptive Moment Estimation for robust convergence

Scheduler
Cosine Learning Rate

Efficiency
Mixed Precision (AMP)
FP16/FP32 hybrid training for accelerated processing.

Guardrails
Early Stopping (Monitor: Val F1)
Patience = 2 epochs, Restore Best Weights

Tracked Metrics: Accuracy, F1-Score, AUC

The Verdict

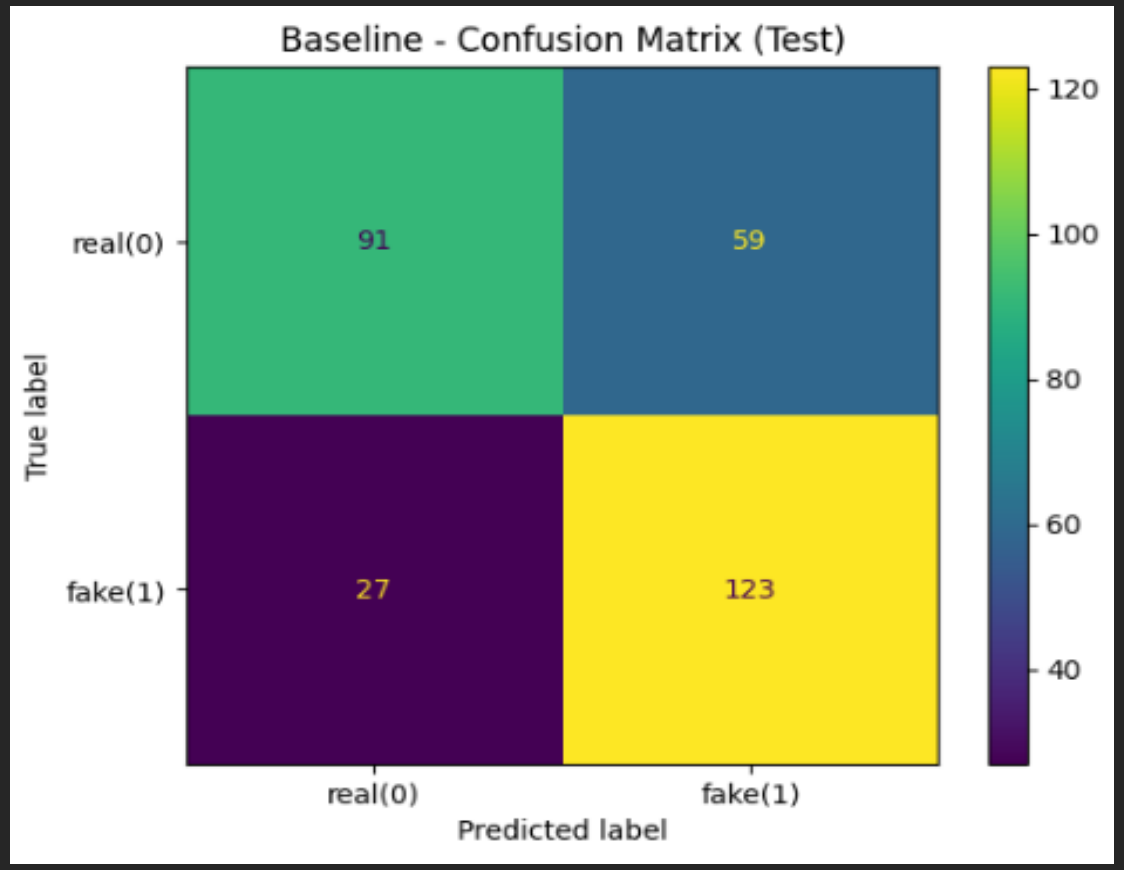
Complexity vs. Efficiency

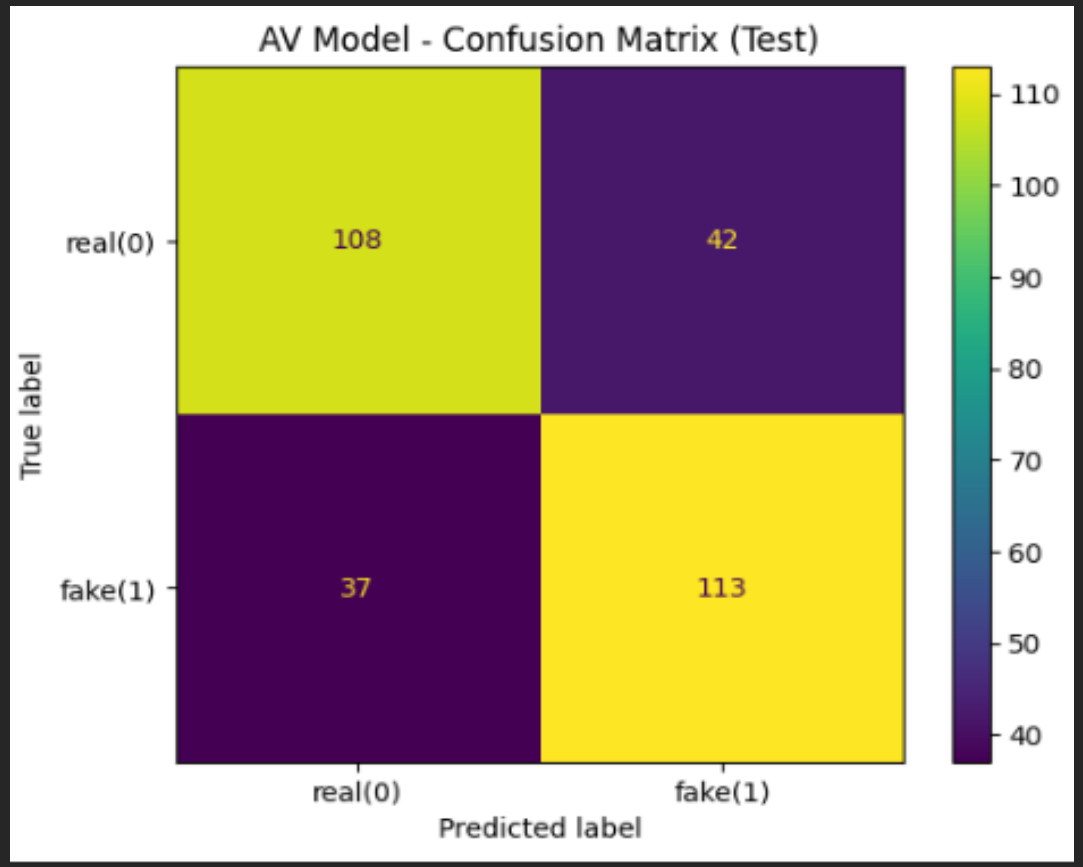
✓ Baseline (Video-Only)
Accuracy: 71.3%
F1-Score: 0.710
AUC: 0.82

✓ AV Model (Multimodal)
Accuracy: 73.7%
F1-Score: 0.737
AUC: 0.81

Key Insight: *AV improves overall accuracy, but introduces a trade-off: FP↓ and FN↑*

Forensic Analysis: Failure Modes

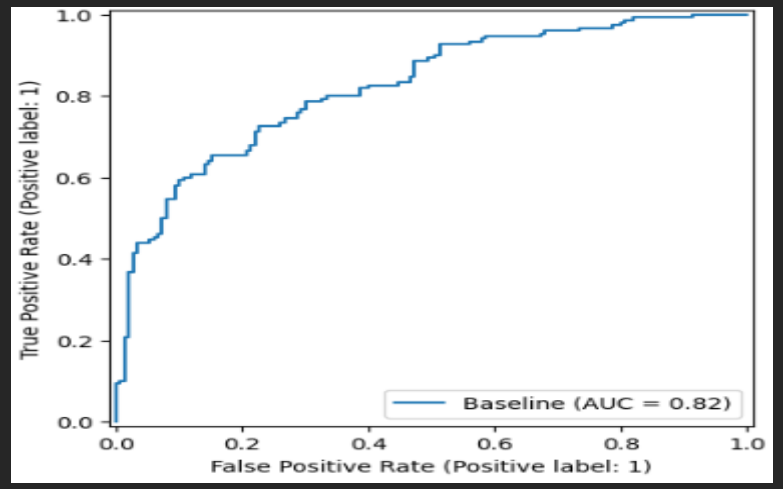




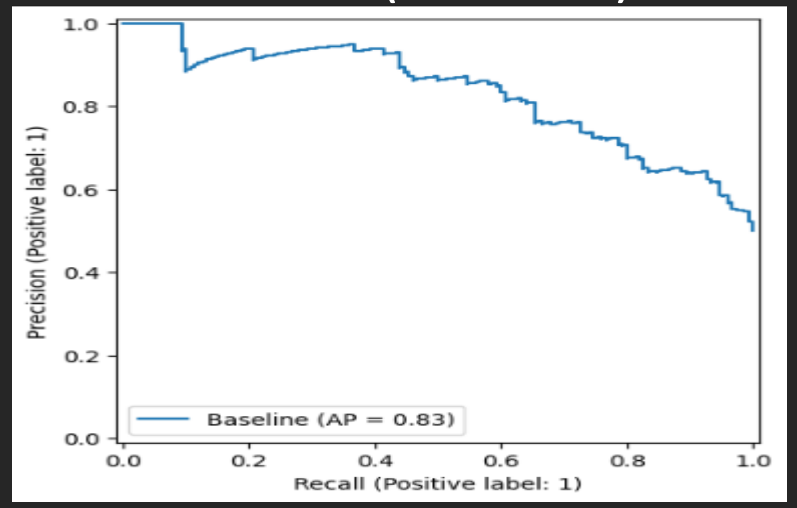
AV model reduces false positives but misses more fakes (FP↓, FN↑)

Stability and Performance Curves

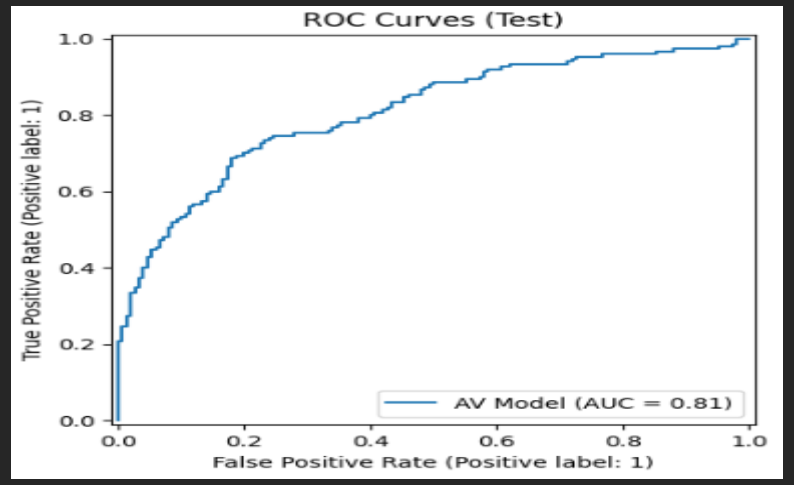
ROC - Baseline (AUC=0.82)



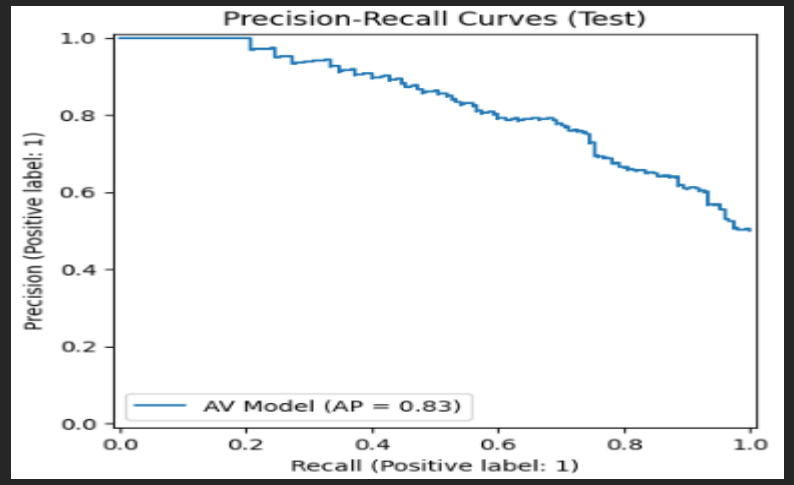
PR - Baseline (AP=0.83)



ROC - AV Model (AUC=0.81)



PR - AV Model (AP=0.83)



AP is equal (0.83), while Baseline has slightly higher AUC (0.82 vs 0.81)

Explainability: Grad-CAM

Validating Feature Focus

CLASS: TRUE REAL | PREDICTION: REAL

METHODOLOGY: 2D Grad-CAM wrapper on 3D input.

OBSERVATION: Model attends to facial features, not background noise.

CONCLUSION: Decision is semantically valid.

Conclusions and Discussion

Visual Dominance:
Baseline AUC=0.82 vs AV AUC=0.81

visuals remain the strongest signal.

XAI Validation:

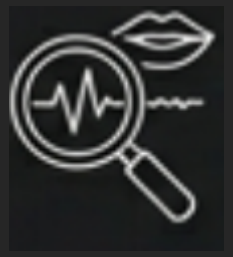
Grad-CAM focuses on facial regions, increasing trust in model decisions.

Multimodal Trade-off:
Accuracy↑

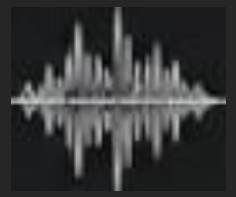
, but FP↓

and FN↑ -fewer false positives, *more missed fakes.*

Future Horizons



Advanced Fusion: SyncNet
Target lip-sync errors specifically rather than general audio features.



Better Audio Representation
Moving beyond MFCC to learn richer spectral features.

Calibration
Threshold tuning to reduce missed fakes (FN↑) without increasing FP too much.

THANK YOU FOR YOUR TIME AND ATTENTION.